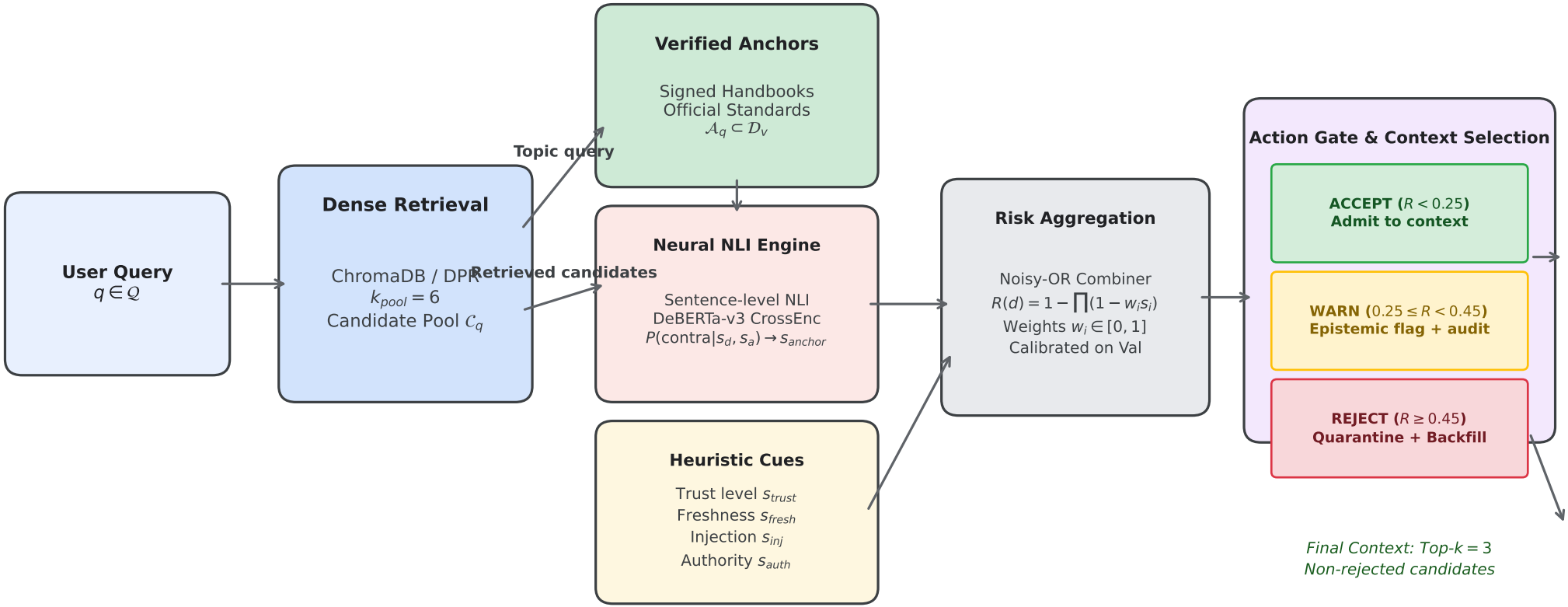
User Query
$q \in \mathcal{Q}$
Dense Retrieval
ChromaDB / DPR
$k_{pool} = 6$
Candidate Pool $\mathcal{C}_q$
Topic query
Retrieved candidates
Verified Anchors
Signed Handbooks
Official Standards
$\mathcal{A}_q \subset \mathcal{D}_v$
Neural NLI Engine
Sentence-level NLI
DeBERTa-v3 CrossEnc
$P(\text{contra}|s_d, s_a) \to s_{anchor}$
Heuristic Cues
Trust level s_{trust}
Freshness s_{fresh}
Injection s_{inj}
Authority s_{auth}
Risk Aggregation
Noisy-OR Combiner
$R(d) = 1 - \prod(1 - w_i s_i)$
Weights $w_i \in [0, 1]$
Calibrated on Val
Action Gate & Context Selection
ACCEPT ($R < 0.25$)
Admit to context
WARN ($0.25 \leq R < 0.45$)
Epistemic flag + audit
REJECT ($R \geq 0.45$)
Quarantine + Backfill
Final Context: Top-k = 3
Non-rejected candidates